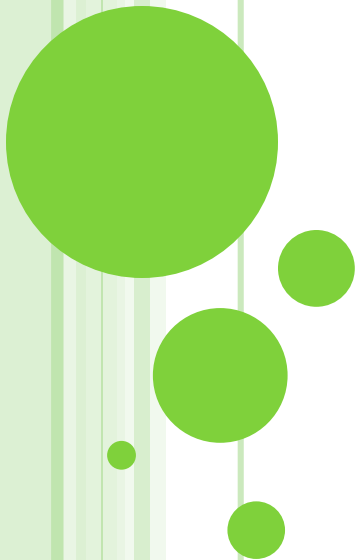


MODULE – II

ER-TO-RELATIONAL MAPPING

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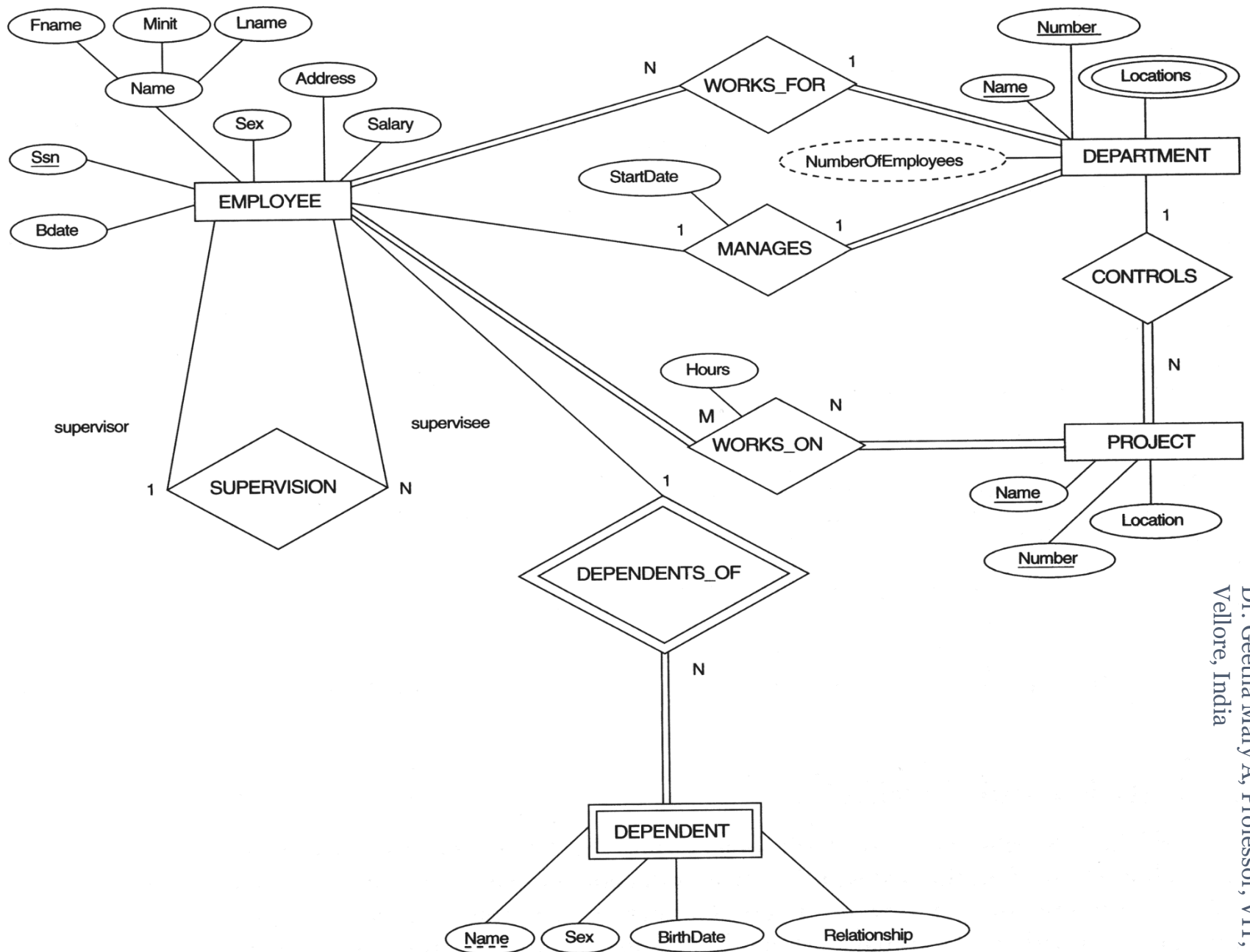
- Database represented by a collection of tables.
- Entity set name → Table name
- Attributes → unique column names



ER-TO-RELATIONAL MAPPING ALGORITHM

- Step 1: Mapping of Regular Entity Types
- Step 2: Mapping of Weak Entity Types
- Step 3: Mapping of Binary 1:1 Relation Types
- Step 4: Mapping of Binary 1:N Relationship Types
- Step 5: Mapping of Binary M:N Relationship Types
- Step 6: Mapping of Multivalued attributes
- Step 7: Mapping of N-ary Relationship Types





STEP 1: MAPPING OF REGULAR ENTITY TYPES.

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
- Choose one of the key attributes of E as the primary key for R. If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.
- **Example:** We create the relations EMPLOYEE, DEPARTMENT, and PROJECT in the relational schema corresponding to the regular entities in the ER diagram. SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT as shown.

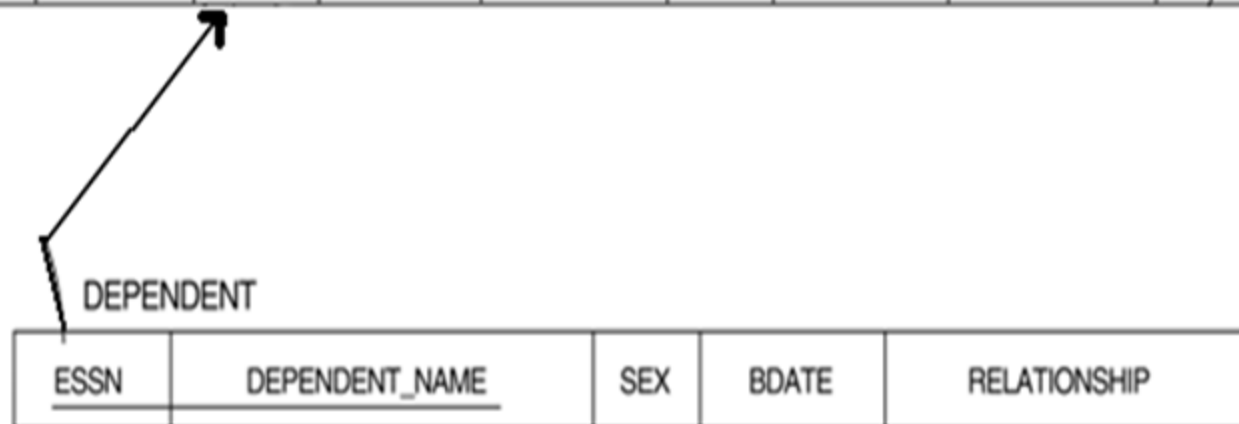


STEP 2: MAPPING OF WEAK ENTITY TYPES

- A weak entity set becomes a table that includes a column for the primary key of the identifying strong entity set
- The primary key of R is the *combination of* the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

EMPLOYEE

FNAME	MINIT	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
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STEP 3: MAPPING OF BINARY 1:1 RELATIONSHIP TYPES

For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R. There are three possible approaches:

(1) Foreign Key approach: Choose one of the relations-S, say-and include as foreign key in S the primary key of T. It is better to choose an entity type with *total participation* in R in the role of S.

Example: 1:1 relationship MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.

(2) Merged relation option: An alternate mapping of a 1:1 relationship type is possible by merging the two entity types and the relationship into a single relation. This may be appropriate when *both participations are total*.

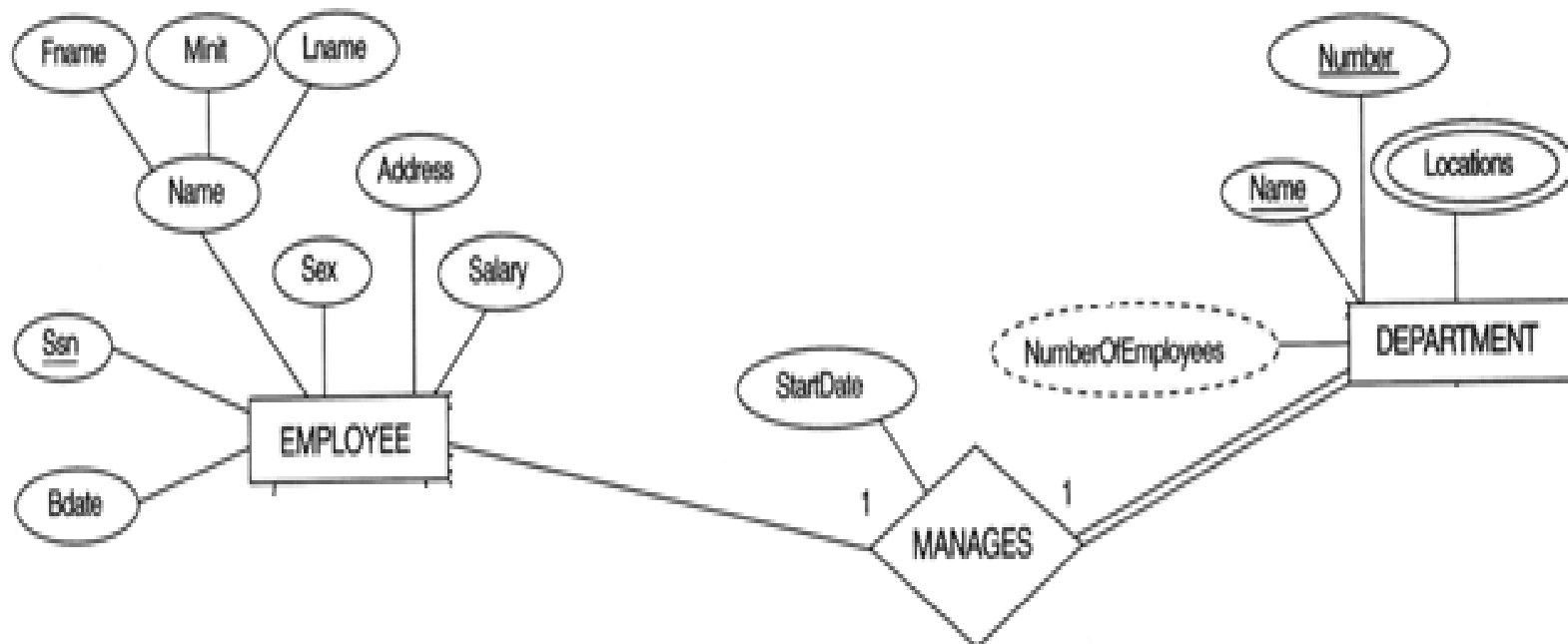
(3) Cross-reference or relationship relation option: The third alternative is to set up a third relation R for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types.

EMPLOYEE

FNAME	MINIT	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
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DEPARTMENT

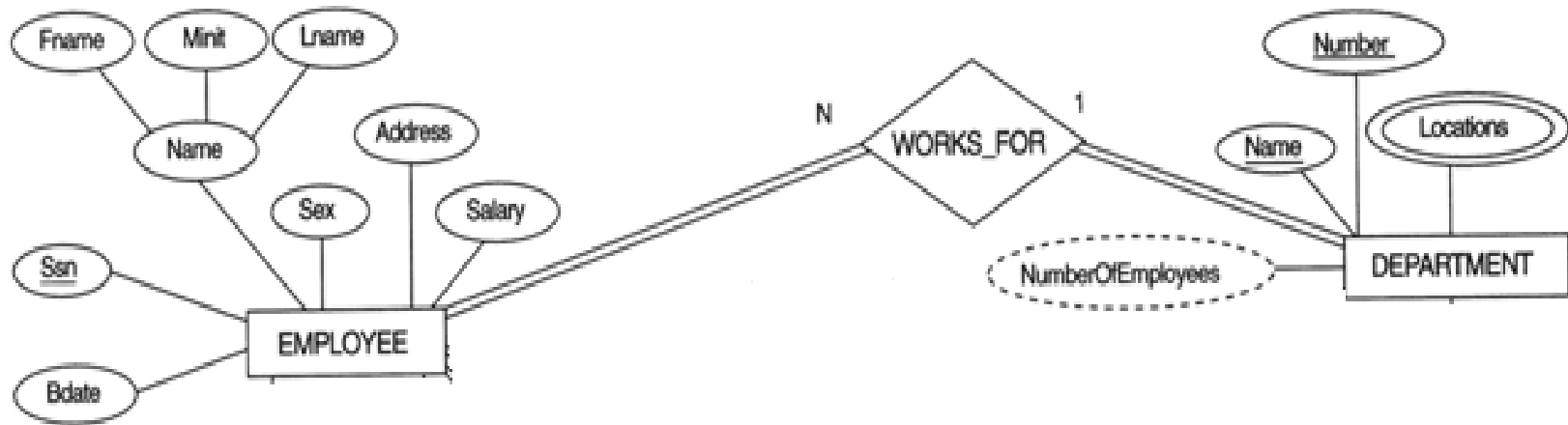
DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE
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STEP 4: MAPPING OF BINARY 1:N RELATIONSHIP TYPES.

- For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
- Include any simple attributes of the 1:N relation type as attributes of S.





EMPLOYEE

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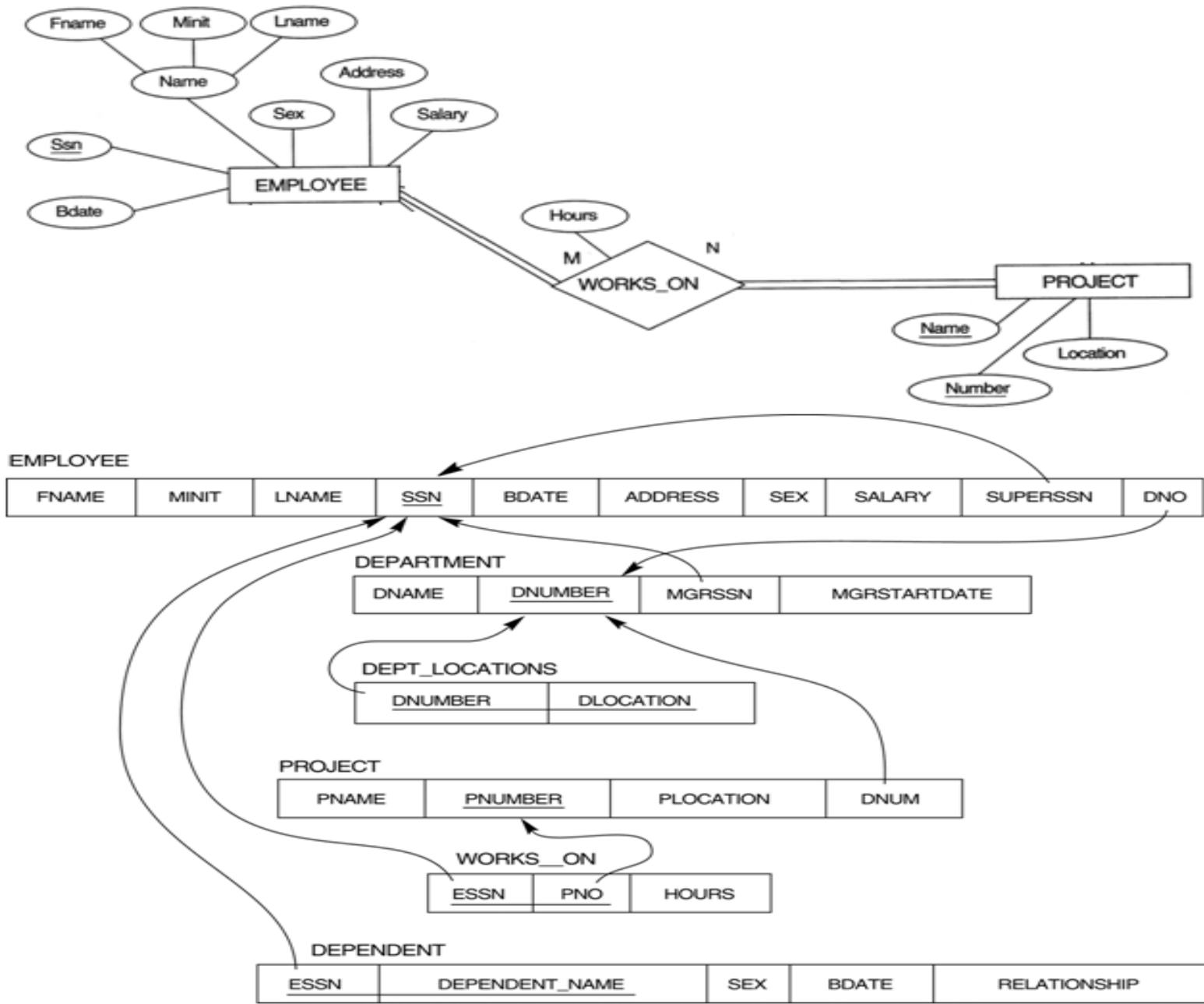
DEPARTMENT

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STEP 5: MAPPING OF BINARY M:N RELATIONSHIP TYPES.

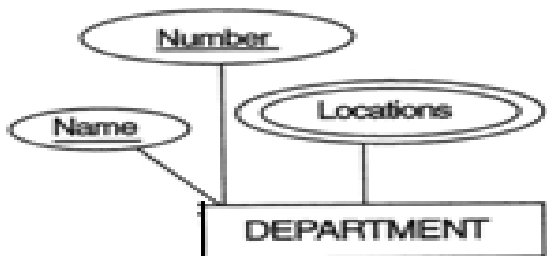
- For each regular binary M:N relationship type R, *create a new relation S* to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.





STEP 6: MAPPING OF MULTIVALUED ATTRIBUTES.

- For each multivalued attribute A, create a new relation R. This relation R will include an attribute corresponding to A, plus the primary key attribute K- as a foreign key in R- of the relation that represents the entity type of relationship type that has A as an attribute.
- The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.



DEPARTMENT

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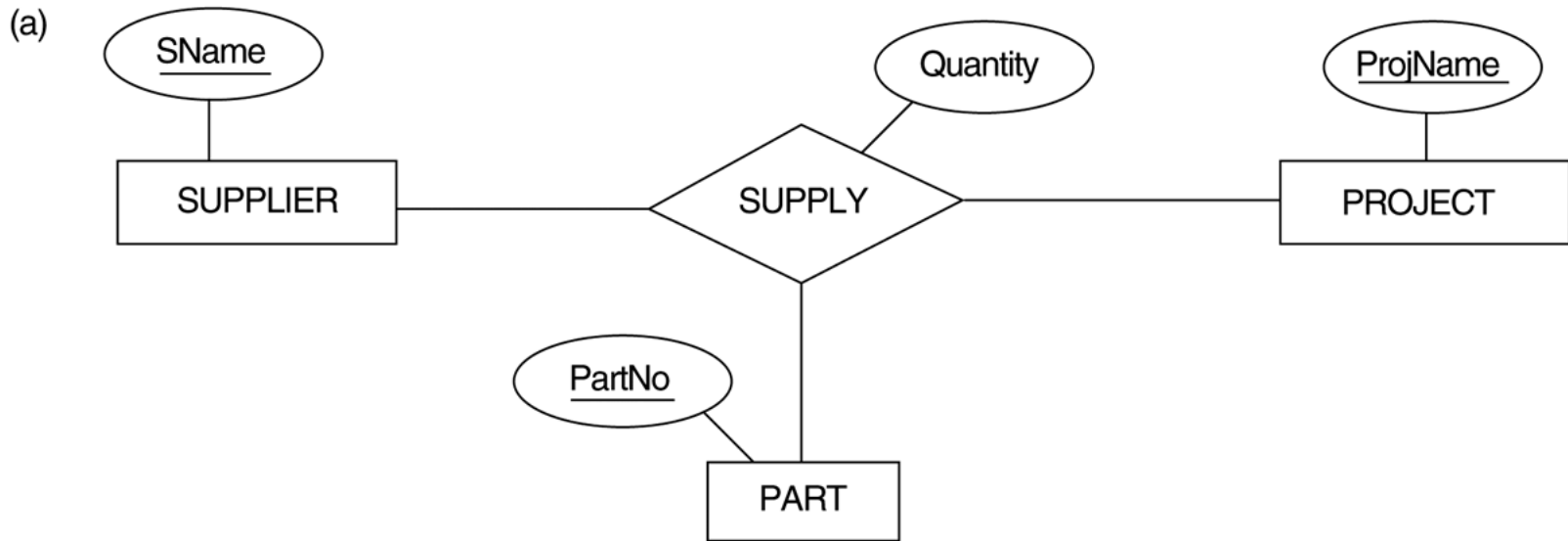
DEPT_LOCATIONS

<u>DNUMBER</u>	<u>DLOCATION</u>
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STEP 7: MAPPING OF N-ARY RELATIONSHIP TYPES.

- For each n-ary relationship type R, where $n > 2$, create a new relationship S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.



PROJECT

<u>PROJNAME</u>	...
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SUPPLIER

<u>SNAME</u>	...
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PART

<u>PARTNO</u>	...
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SUPPLY

<u>SNAME</u>	PROJNAME	<u>PARTNO</u>	QUANTITY
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